

Foundational Research & Advances in Quantum Reinforcement Learning (QRL)

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1. ABSTRACT & THEORETICAL FOUNDATION

This peer-reviewed reference document synthesizes the formal mathematical formulation, operational circuit protocols, and physical implementation benchmarks for Quantum Reinforcement Learning (QRL) in modern quantum information architectures.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"A video game player that explores every possible maze path at once in a ghost-like superposition, discovering the winning strategy quadratically faster."

3. Mathematical Formulation & Unitary Rule

$$Q(s, a) = \langle \theta | U^\dagger(s, a, \vec{\theta}) M U(s, a, \vec{\theta}) | \theta \rangle, \quad \text{quad}$$

Grover-like amplitude amplification quadratically speeds up the search for optimal actions in state s .

5. Executable IBM Qiskit (Python) Code

```
from qiskit.circuit import QuantumCircuit, Parameter

# State-action encoding circuit for quantum policy gradient
qc = QuantumCircuit(2)
state_param = Parameter('state')
action_param = Parameter('action_weight')

qc.rx(state_param, 0)
qc.ry(action_param, 1)
```

6. Computational Complexity & Speedup

Classical Time:

Quantum Time:

Speedup Class:

$O(|S| * |A|)$ exploration complexity (is) $O(\sqrt{|S| * |A|})$ quantum query complexity to state-action oracle

7. Key Applications & Future Research Directions

- Autonomous drone navigation and path planning in dynamic environments
- High-frequency algorithmic trading policy optimization
- Automated quantum circuit design and pulse calibration agents

Future Work:

Hardware fidelity improvements, compiler transpilation optimizations, and integration with fault-tolerant MachineLearning pipelines.